



Pharm*Achieve*

ERECTILE DYSFUNCTION

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

BP Blood Pressure

BPH Benign Prostatic Hyperplasia

CBC Complete Blood Count

cGMP Cyclic Guanosine Monophosphate

ED Erectile Dysfunction

GnRH Gonadotropin-Releasing Hormone

LUTS Lower Urinary Tract Symptoms

MAOI Monoamine Oxidase Inhibitor

NPT Nocturnal Penile Tumescence

PDE5 Phosphodiesterase Type 5

PRN When Needed

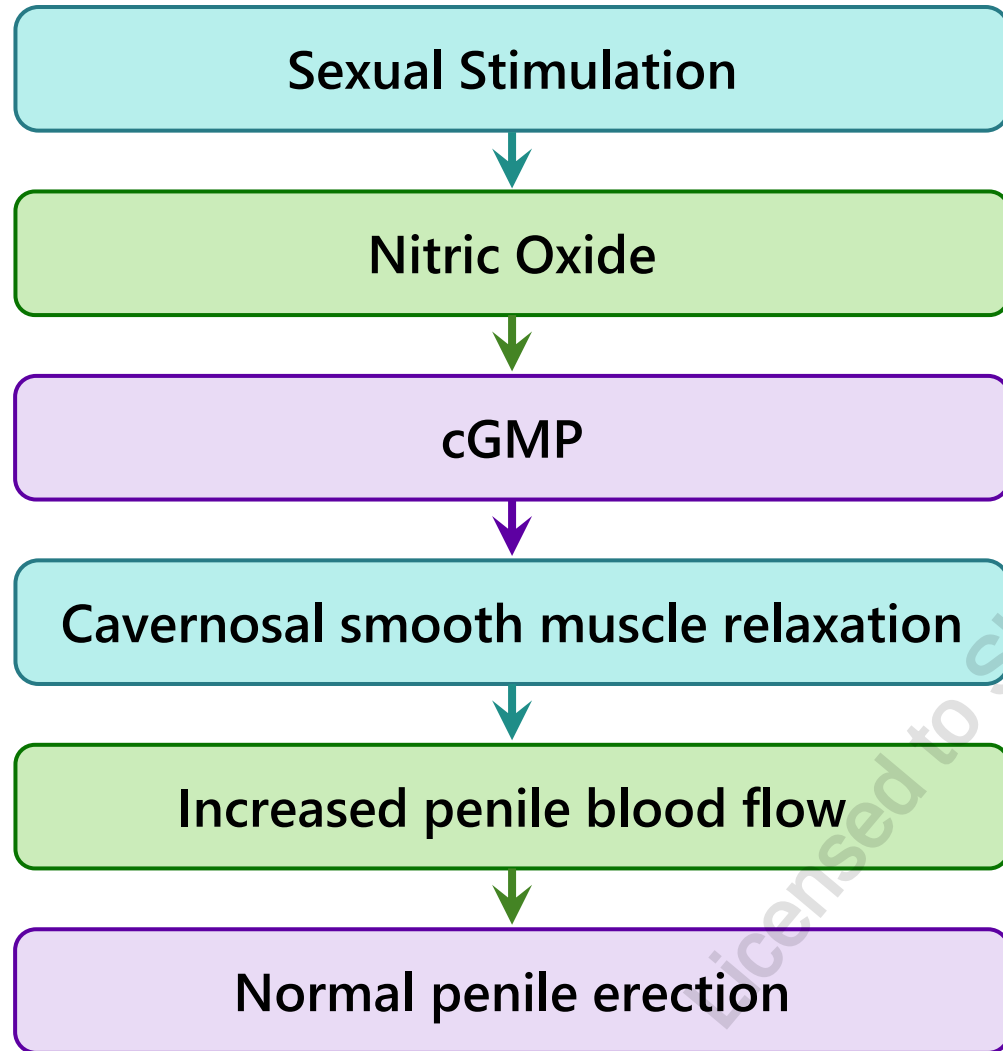
SSRI Selective Serotonin Reuptake Inhibitor

TCA Tricyclic Antidepressant

OBJECTIVES

- 1 Identify pathophysiology, signs and symptoms, and risk factors for erectile dysfunction (ED)
- 2 Identify drugs and medical conditions that can cause ED
- 3 Review available treatment options for ED

PHYSIOLOGY OF NORMAL ERECTILE FUNCTION



PATHOPHYSIOLOGY

 **Erectile dysfunction (ED)**, also known as **impotence**, is defined by the “**inability to achieve or sustain an erection for satisfactory sexual activity**”

Erectile dysfunction is **different from other conditions** that interfere with male sexual intercourse, such as:

- Lack of sexual desire (decreased libido)
- Problems with ejaculation and orgasm (ejaculatory dysfunction)

Sexual stimulation is required for normal erectile function

- Neural impulses are conveyed along the spinal cord, exiting through the parasympathetic preganglionic nerves and their **nerve endings release nitric oxide**
- Nitric oxide stimulates the enzyme guanylyl cyclase to produce the intracellular mediator **cyclic guanosine monophosphate (cGMP)**
- Combination of:
 1. **Continuous blood flow into the penis**
 2. **Limited blood outflow from the penis** } **Increased intracavernosal pressure** at which the penis has sufficient rigidity to permit vaginal penetration

CLINICAL PRESENTATION

- ED is **highly prevalent in aging men**
- Often causes serious distress, prompting men to seek medical attention they may not otherwise seek
- Has a profound effect on:
 - **Intimate relationships**
 - **Quality of life**
 - **Overall self-esteem**
- ED may also be the **presenting symptom of undetected cardiovascular disease (remains controversial)**
- **Economic impact of ED** is multifactorial that include:
 - **Direct costs**
 - Physician evaluation, pharmacotherapy & diagnostic testing
 - **Indirect costs**
 - Lost time at work, lost productivity, effects on the man's partner, family, & coworkers

RISK FACTORS

- Advancing age
- Diabetes
- Dyslipidemia
- Hypertension
- Obesity
- Metabolic syndrome
- Sedentary lifestyle
- Cigarette smoking
- Penile trauma (Peyronie's disease)
- Pelvic, penile, urethral surgery
- Prostatectomy and prostate radiotherapy
- LUTS/BPH
- Illicit drug use (e.g. cocaine, methamphetamine)
- Psychological conditions (e.g. anxiety, depression, guilt, history of sexual abuse, marital or relationship problems, stress)
- Drug causes

DRUG CAUSES

- Opioids
- 5-alpha reductase inhibitors
- SSRIs, TCAs, MAOIs
- Lithium, phenothiazines
- Acute and/or chronic alcohol use
- Carbamazepine, phenytoin, barbiturates
- Spironolactone, thiazide diuretics, acetazolamide
- Digoxin
- Alpha-blockers, clonidine
- Beta-blockers
- Anti-androgen, GnRH analogues

DIAGNOSIS

Detailed patient history	Medical and psychological comorbidities, medications, substance use history, surgical/radiation history, previous treatments
Sexual Health Inventory for Men (SHIM)	Validated assessment tool useful in initial diagnosis and follow-up of ED (patient-reported assessment of ED)
Physical exam	Overall: BP, mood, gynecomastia; penis and groin: penile length/girth, penile plaques, quality of femoral pulses; testicles: volume and consistency
Laboratory investigations	Baseline A1C, fasting glucose, and lipid profile, CBC, liver function tests, testosterone level
NPT testing- FYI	Minimally invasive procedure to measure and record nighttime erectile events, limited availability in Canada, cost not covered in most provinces (limited situations by specialists in sexual medicine)

GOALS OF THERAPY

- ✓ Treat underlying conditions and reversible causes
- ✓ Address any contributing emotional factors
- ✓ Restore erectile capacity so that it is sufficient for predictable sexual intercourse
- ✓ Optimize non-pharmacological and pharmacological interventions to treat erectile dysfunction
- ✓ Try to tailor the treatment to the patient's personal preferences
- ✓ Improve quality of relationship between patient and sexual partner

NON-PHARMACOLOGICAL MEASURES

Counseling

- Psychological factors
- Partner issues
- Economic factors



Treatment of comorbidities

- Optimize management of comorbidities (e.g. diabetes, hypertension, dyslipidemia, depression)



Lifestyle modifications

- Weight loss
- Regular physical activity (aerobic and/or resistance training)
- Smoking cessation
- Tighter glycemic control (dietary restrictions)
- Reduce alcohol intake (can improve progression over time)



Vacuum Erection Devices

- Can be maintained safely for up to 30 minutes
- No absolute contraindications
- May cause bruising, numbness, pain

PHARMACOLOGICAL MEASURES

PHOSPHODIESTERASE TYPE 5 INHIBITORS

Drugs	Sildenafil, vardenafil, tadalafil
MOA	Inhibition of PDE5 (enzyme which degrades cGMP) located in penile vascular smooth muscle cells. cGMP promotes smooth muscle relaxation in response to sexual stimulation
Duration	PRN use or daily use (only with tadalafil)
Interactions	Nitrates (regularly or intermittently), antihypertensives (increased risk of hypotension) PDE5i are metabolized by CYP3A4 (be mindful of strong inhibitors: ritonavir, erythromycin), cimetidine (only with sildenafil). Caution with alpha blockers (separate dosing)
AEs	Headache, flushing, dyspepsia, nasal stuffiness (universal in all three) Visual symptoms-blue vision, blurred vision, sensitivity to light (sildenafil, vardenafil) Myalgias (tadalafil) QTc prolongation (vardenafil) Rare: syncope, and nonarteritic anterior optic neuropathy
Tips	1st line therapy for most patients, Requires Sexual Stimulation Similar efficacy within class. Overall efficacy depends on mental arousal Do not use nitrates concomitantly due to risk of serious/fatal decrease in blood pressure If nitrate therapy is required, it can be used 24h after sildenafil & vardenafil, but 48h after tadalafil use; close BP monitoring is required

PHARMACOLOGICAL MEASURES

	Sildenafil	Vardenafil	Tadalafil
Onset of action/Time to peak effect	Onset: 30-60 minutes (as early as 15 minutes) Time to peak: 1 hour	Onset: 30-60 minutes (as early as 10 minutes) Time to peak: 1 hour	Onset: 60 minutes Time to peak: 2 hours
Half-life	4 hours	4 hours	17.5 hours
Duration	Up to 8-12 hours	Up to 8-12 hours	Up to 36 hours
Food interaction	Avoid with fatty meal	Avoid with fatty meal	No effect
Dosing	50*-100 mg PRN Max 100 mg/day *may start at a lower dose (25 mg)	5-20 mg PRN max 20 mg/day	PRN: 10-20 mg; Max 20 mg/36-48h CrCl < 30 mL/min: 5 mg q72h or Daily dosing: 2.5-5 mg PO daily (especially if LUTS present)
Dosing Adjustment	Renal, Hepatic 65+ years old CYP3A4i	Hepatic 65+ years old CYP3A4i	Renal, Hepatic 65+ years old CYP3A4i CrCl < 30 mL/min; Severe hepatic impairment: use not recommended
Unique A.E.	Visual Symptoms	Visual Symptoms QTc Prolongation	Myalgias



Switch to another PDE5 inhibitor (if tried first one 8 times and was unsuccessful)
or add testosterone to improve ED or reassess how PDE5i is used

PHARMACOLOGICAL MEASURES

PROSTAGLANDIN E1 ANALOGUES

* MUSE® currently has a 'dormant' status on the drug product database

Drugs	Alprostadil: Caverject® (intracavernosal injection) or MUSE®*(intraurethral pellet)
MOA	Prostaglandin E1 is a vasodilator that relaxes smooth muscle thereby allowing increased blood flow within the blood vessel
Directions for Use & Duration	CAVERJECT®: use PRN 5-10 minutes before sexual activity, max once daily or 3 times per week with at least 24 hours between doses, duration up to 60 minutes MUSE®: use PRN 10-30 minutes before sexual activity, max twice/24 hours, duration 30-60 minutes
Interactions	PDE5i
AEs	CAVERJECT®: penile pain, dizziness, headache, priapism (>4 hours, 4%), bruising/hematoma at injection site MUSE®: penile pain, urethral burning, dizziness, headache, testicular pain, vulvovaginal pruritis (female partner)
Tips	• 2nd line option • CAVERJECT: effective; generally well tolerated; teaching session advised for technique and appropriate dose titration • MUSE: select patients; suboptimal efficacy; urethral discomfort; dose titration with in-office trial; preferred over CAVERJECT as does not require injection • Patients should be informed about the potential for occurrence of prolonged erections and should seek emergent medical evaluation for rigid erections lasting longer than four hours (priapism) • Contraindications: anticoagulation (intracavernosal injection only), history of priapism, conditions that increase the risk of priapism (e.g. leukemia, sickle cell disease)

PHARMACOLOGICAL MEASURES

Androgens

Drugs	Testosterone (patch, gel, tablets, or IM injections)
MOA	Testosterone replacement therapy
Time to peak	IM median = 7 days, TD median = 8 hours, PO median = 4-5 hours
Interactions	Agents with blood glucose lowering effects, propranolol, corticosteroids, vitamin K antagonists (i.e. warfarin)
AEs	Acne or oily skin, mild fluid retention, stimulation of prostate tissue, breast enlargement, worsening of sleep apnea, decreased testicular size, hair growth
Tips	Not as monotherapy – may be used in combination with a PDE5i Inadequate production of testosterone is not a common cause of ED, but testosterone may help improve the ED Topical gel has lower risk of skin reaction than patch Gel: Avoid swimming, showering or washing the application site for at least 5 hours following application. If pregnant or considering becoming pregnant, family members/spouses should avoid coming into contact with the product

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
Erectile function (quality, duration)	Improved	While on therapy (onset and duration of action is medication specific)
Patient and partner overall satisfaction		

Safety Endpoints		
Parameter	Degree of Change	Timeframe
Blood pressure (especially with concomitant use of antihypertensives)	Maintain blood pressure > 90/60 mmHg	Duration of therapy
AEs e.g. visual disturbances (sildenafil and vardenafil), myalgias (tadalafil), headache, flushing	Prevent/minimize	
Priapism	Prevent	

TAKEAWAY

- Pharmacists must prioritize privacy, respect and discretion when discussing sexual dysfunction
- **First-line therapy:** PDE5 inhibitors. They are metabolized by CYP 3A4 and interact with anti-hypertensives and nitrates, among others
- Patient should be counseled to take the medication about 30-60 minutes before anticipated sexual encounter
- Tadalafil – certain patient-centered factors influence daily versus on-demand dosing (**concomitant LUTS, cost, female partner preference**)
- **Second-line therapy:** Prostaglandin E1 analogues
- **Androgens should not be used as monotherapy**
- Managing comorbidities and lifestyle modifications are important non-pharmacological measures
- **Priapism: erection lasting > 4 hours → urgent referral**

CASE

KB is a 63-year-old male who presents to his family physician expressing concerns of erectile dysfunction. He has started dating again but has been avoiding being intimate with his partner due to difficulty maintaining an erection in his previous relationship. He feels embarrassed and is seeking your advice. KB has hypertension for which he takes amlodipine 5 mg po daily. Additionally, he takes saw palmetto 450 mg once daily, vitamin C 500 mg once daily, and vitamin D 2000 units once daily. He has a sulfa allergy resulting in a rash. He eats takeout food (e.g. burgers, pizza, fries) most days.

1. You counsel KB on lifestyle factors but he is a busy individual who does not want to change his diet. What would be the best recommendation for him?
 - a) Sildenafil
 - b) Vardenafil
 - c) Tadalafil
 - d) Alprostadil injection



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2. KB is concerned about side effects of medication use. Which of the following statements about phosphodiesterase type 5 inhibitors is incorrect?
- a) All the drugs within this class can cause headaches, dyspepsia, and flushing
 - b) Sildenafil can cause visual symptoms
 - c) Tadalafil can cause myalgias
 - d) Tadalafil can result in QTc prolongation



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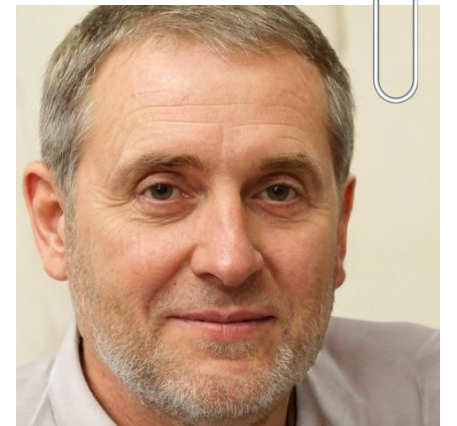
3. KB asks if there is anything else he needs to know when it comes to his medication. Which of the following statements is correct?
- a) Tadalafil does not need to be adjusted for hepatic dysfunction
 - b) Tadalafil dose needs to be decreased for KB given his age
 - c) There is a significant interaction between PDE5 inhibitors and nitrates which can cause a hypertensive crisis
 - d) KB must seek immediate medical attention if he experiences priapism



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4. KB has been on tadalafil 5 mg once daily as needed for a few months now. His MD calls him to inform him that there has been a significant decline in his kidney function. What is the most appropriate course of action?
- a) Reduce dose to 2.5 mg once daily
 - b) Stop the medication immediately as it is not recommended in renal dysfunction
 - c) Switch the dosing to 5 mg every other day
 - d) Switch to vardenafil



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CHANGE LOG

August 2025

- Slide 11: edited to reduce alcohol intake

September 2025

- Slide 9: Removed 'FYI'

November 2025

- Added diagram illustrating the physiology of erectile function.
- Slide 12: Added treating comorbidities (diabetes, hypertension, dyslipidemia), added weight loss, and aerobic and/or resistance training
- Slide 13: Added caution with alpha-blockers under interactions and noted the need for separated dosing.
- Slide 14: Changed sildenafil dosing to 25–100 mg PRN; added "PRN." Added "especially if LUTS is present" under tadalafil dosing.
- Slide 15: Added a note that MUSE® currently has a 'dormant' status on the drug product database
- Slide 17: Added treatment efficacy, onset/duration expectations, additional side effects, blood pressure monitoring, and administration technique.

March 2026

- Slide 6: Minor update to definition
- Slide 10: Added 'FYI' to NPT testing
- Slide 12: Added some details re: vacuum erection devices
- Slide 14: Removed some brand names as they were cancelled post-market
- Slide 19: Changed option d) to alprostadil injection
- Slide 21: Changed options for question 3 for clarity